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CO4 – ASSIGNMENT 4

SEMANTIC ANALYSIS AND SEMANTIC SYSTEMS ENGINEERING

Course Code: DSA03

Course Title: Natural Language Processing

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ASSIGNMENT 4

Meaning Representation, Predicate Logic, Word Sense Disambiguation and Semantic Retrieval

This report answers Assignment 4 Parts I and II. It represents event meaning, formalises first-order statements, analyses attachment ambiguity, designs and tests word-sense disambiguation, and traces semantic retrieval decisions. Every formula defines its vocabulary before use; alternative interpretations are retained where the input alone is insufficient.

Contents

- Part I – Tasks 1 to 6: Semantic Analysis
- Part II – Tasks 1 to 6: Semantic Systems Engineering
- References

Assumptions. Names denote distinct individuals unless stated otherwise. Event variables range over events. Retrieval scores are deterministic illustrative values used to make the ranking rule reproducible.

PART I

Task 1 – Meaning Representation and Language Structure (8 Marks)

Input: “Maya gave Ravi a book yesterday.”

1.1 Meaning components

Component	Value	Role / observation
Entities	Maya; Ravi; book b	Maya and Ravi are individuals; “a book” introduces b.
Event	giving event e	Transfer-of-possession event.
Participants	Maya, Ravi, b	Agent/giver, recipient/goal, theme.
Time	yesterday	Locates e before the utterance day.
Surface versus meaning	S V IO DO Adv	Word order signals a ditransitive; meaning is the role structure.

1.2 Two representations

Event structure: GIVE(e) ∧ Agent(e,Maya) ∧ Theme(e,b) ∧ Book(b) ∧ Recipient(e,Ravi) ∧ Time(e,yesterday). The transfer outcome, Owns(Ravi,b), is a defeasible inference rather than literal content.

Frame slot	Value
Event type	GIVE
Agent	Maya
Theme	book b
Recipient	Ravi
Time	yesterday

Implicit	exact time, voluntariness and book identity are not stated

The formula exposes variables and truth conditions; the frame is easy to query or serialise. Both leave unstated whether Ravi accepted the book.

1.3 Meaning structure and ambiguity

English subject position supports Agent and the first object supports Recipient. “Yesterday, Maya gave a book to Ravi” has the equivalent core representation because Agent, Theme, Recipient and Time retain the same values. The preposition “to” marks Recipient explicitly.

1.4 Design defence

A computer can query named roles, link later facts to e or b, and validate that a GIVE event has a recipient. A paraphrase alone cannot reliably support such operations because it does not label the relationships or separate the event from its participants.

Task 2 – First Order Predicate Calculus (8 Marks)

2.1 Vocabulary and 2.2 translation

Symbol	Meaning
Student(x), Course(c), Instructor(i)	x/c/i have the indicated type
Enrolled(x,c)	student x is enrolled in c
Teaches(i,c)	instructor i teaches c
Passed(x,c)	student x passed c
rao	constant for Dr. Rao

- (i) $\forall x\,[\text{Student}(x) \rightarrow \exists c\,(\text{Course}(c) \wedge \text{Enrolled}(x,c))]$.
- (ii) $\exists x\,[\text{Student}(x) \wedge \forall c\,((\text{Course}(c) \wedge \text{Teaches}(\text{rao},c)) \rightarrow \text{Passed}(x,c))]$.
- (iii) $\neg\exists i\,[\text{Instructor}(i) \wedge \forall c\,(\text{Course}(c) \rightarrow \text{Teaches}(i,c))]$.

2.3 Scope and counterexample

In (ii), the same student must pass every Rao-taught course. Reversing the order gives $\forall c[(\text{Course}(c) \wedge \text{Teaches}(\text{rao},c)) \rightarrow \exists x(\text{Student}(x) \wedge \text{Passed}(x,c))]$, which may use a different student per course.

Finite model	∃x∀c result	∀c∃x result
Rao teaches NLP and IR; Ana passed NLP only; Bala passed IR only	False: no one passed both.	True: Ana witnesses NLP and Bala witnesses IR.

2.4 Consistency

$\forall x[\text{Student}(x) \rightarrow \text{Passed}(x,\text{nlp})]$ and $\exists x[\text{Student}(x) \wedge \neg\text{Passed}(x,\text{nlp})]$ cannot both be true if NLP and Passed have the same interpretation and assessment period.

Task 3 – Syntax Driven Semantic Analysis (9 Marks)

Input: “The experienced researcher saw the student with the telescope.”

Concept	Role
researcher r	entity; Agent/experiencer
student s	entity; Theme
telescope t	entity; instrument or student modifier

experienced(r)	modifier of researcher
see(e)	perception event
with	ambiguous PP relation

3.2–3.3 Attachment and composition

Parse	Representation	Interpretation
VP attachment	See(e) ∧ Agent(e,r) ∧ Theme(e,s) ∧ Instrument(e,t)	Researcher used the telescope.
NP attachment	See(e) ∧ Agent(e,r) ∧ Theme(e,s) ∧ With(s,t)	Researcher saw a telescope-carrying student.

Composition builds noun predicates, intersects Experienced with Researcher, composes see with its agent and theme, and then attaches the PP. The readings diverge exactly when “with the telescope” becomes a VP adjunct or an NP modifier.

3.4 Failure analysis

Error	Effect	Check
Wrong VP attachment	Claims the researcher used a telescope.	Compare discourse context and Instrument selectional preference.
Roles from word order only	Fails on passive or topicalised inputs.	Use dependency relations and verb valency, not position alone.

Task 4 – Lexemes Senses and WSD (9 Marks)

Context	Word form / lexeme	Sense / POS
deposited cash at bank	bank; BANK noun	financial institution
bank of the river	bank; BANK noun	river edge
pilot banked aircraft	banked; BANK verb	tilt aircraft in a turn

A word form is the observed token; a lexeme groups related forms/senses. The noun and verb entries differ grammatically even though spelling overlaps.

4.2–4.3 Reproducible WSD method

Input: the target, POS tag, dependency neighbours and ±4 tokens. Candidates: FINANCIAL_NOUN, RIVER_EDGE_NOUN, AIRCRAFT_TILT_VERB. Feature rules: {cash, deposit, account, loan}=+3 financial; {river, water, sat, shore}=+3 river; {pilot, aircraft, left, turn}=+3 aircraft. POS mismatch=-5. Return highest score; ties retain candidates and request context.

Sentence	Evidence	Output
Nila deposited cash at the bank	deposited + cash	FINANCIAL_NOUN, 6
They sat on the bank of the river	sat + river	RIVER_EDGE_NOUN, 6
Pilot banked aircraft left	pilot + aircraft + left	AIRCRAFT_TILT_VERB, 9
My bank approved the loan	approved + loan	FINANCIAL_NOUN, 6

Failure: “The bank was steep” lacks enough context; use lower confidence rather than an unjustified single label.

4.4 Evaluation

Use 12 labelled items, four per sense. If ten are correct, accuracy=10/12=83.3%. A balanced-set score may not reflect real text, where financial uses are more common and underspecified uses are harder; also report per-sense recall and a confusion matrix.

Task 5 – Information Retrieval and Semantic Matching (8 Marks)

Query intent: LEARN_WSD_METHOD. Concepts={WSD:1.0, context:0.8, ambiguity:0.8, method:0.6}; documents store weighted concepts, embedding and task type.

Keyword matching may miss “Contextual disambiguation of polysemous terms using neighbouring words.” A semantic representation maps polysemous/contextual disambiguation to the WSD intent.

Score(d)=0.45 normalised BM25 + 0.35 cosine(query,document) + 0.20 sense coverage. Break ties by sense coverage, then stable document ID.

At top 5, retrieved={D1,D2,D3,D4,D5}; relevant={D1,D2,D6,D7}; relevant retrieved={D1,D2}. Precision@5=2/5=0.40 and recall@5=2/4=0.50. A related-but-non-answering parsing-ambiguity article is a semantic false positive; add a WSD task-type constraint.

Task 6 – Integrated Semantic Analysis Challenge (8 Marks)

Input: “I saw her duck near the bank, and it was closed.”

Candidate	Senses and representation
A: bird + financial bank	DuckBird(d) ∧ See(e,Speaker,d) ∧ Near(d,b) ∧ BankFinancial(b) ∧ Closed(b). “it”=b.
B: duck action + river bank	DuckAction(e2,Her) ∧ See(e1,Speaker,e2) ∧ Near(e2,b) ∧ BankRiver(b) ∧ Closed(b). “it”=b.

Syntax distinguishes possessive “her duck” from object-pronoun plus verb. Branch/account/opening-hours cues favour financial bank; water/shore/birds favour river. Closed strongly selects an organisation, so A is preferred, but unresolved input should trigger a clarification question.

Document	Match	Score
D1 Bank branch closed hours	financial bank + closed	0.89
D2 Duck species near wetlands	bird + river	0.61
D3 Riverbank erosion	river bank only	0.35
D4 Aircraft banking guide	wrong sense	0.08

A misleading “The river was high” can pull toward B. Refinement: make coreference-compatible selectional preference for Closed favour venues/organisations over landforms.

PART II – Semantic Systems Engineering

Task 1 – Representing Meaning and Language Structure (8 Marks)

Input: “Leena sent the report to Arun after the meeting.”

Role / node	Value
Event e	SEND
Agent	Leena
Theme	report r
Recipient	Arun
Temporal relation	After(e,m), meeting m
Implicit	channel and confirmed receipt are not asserted

Event formula: SEND(e) ∧ Agent(e,Leena) ∧ Theme(e,r) ∧ Report(r) ∧ Recipient(e,Arun) ∧ Meeting(m) ∧ After(e,m). Graph slots: e:SEND →agent Leena; →theme r:REPORT; →recipient Arun; →after m:MEETING. “After the meeting, Leena sent Arun the report” preserves all core relations. “Leena sent the meeting report” changes meeting from a prior event to a report modifier.

Validation rule: SEND must have exactly one Agent, Theme and Recipient; flag missing roles or a recipient type incompatible with the send lexicon.

Task 2 – FOPC Formalise and Test (9 Marks)

Symbol	Meaning
Learner(x), Course(c)	type predicates
Enrolled(x,c), Completed(x,c)	enrolment and completion
Prerequisite(p,c)	p is prerequisite of c
nlp, dept	constants

- (i) ∀x [Learner(x) ∧ Enrolled(x,nlp) → ∃p (Course(p) ∧ Prerequisite(p,nlp) ∧ Completed(x,p))].
- (ii) ∃x [Learner(x) ∧ ∀c ((Course(c) ∧ OfferedBy(c,dept)) → Completed(x,c))].
- (iii) ¬∃x∃c [Learner(x) ∧ Course(c) ∧ Completed(x,c) ∧ ¬Enrolled(x,c)].

Swapping ∃x∀c in (ii) to ∀c∃x changes the claim: with Ana completing NLP and Bala completing IR, each course has a completer but no learner completes every course. They are not equivalent.

KB item	Content
Facts	Learner(ana); Enrolled(ana,nlp); Prerequisite(py,nlp); Completed(ana,py)
Rule	Learner(x) ∧ Enrolled(x,nlp) ∧ Prerequisite(p,nlp) ∧ Completed(x,p) → Eligible(x,nlp)
Query	Eligible(ana,nlp) follows by x=ana,p=py.

Task 3 – Linguistic Concepts and Syntax Driven Analysis (9 Marks)

Input: “The analyst noticed the student with the camera near the library.”

Attachment	Meaning
NP attachment	Notice(e,a,s) ∧ With(s,camera) ∧ Near(s,library): the student has the camera.
VP attachment	Notice(e,a,s) ∧ Instrument(e,camera) ∧ Near(e,library): analyst

	used camera near library.

Build Analyst(a), Student(s), Camera(c), Library(l), then combine constituents. The composition diverges when the PP is attached under NP rather than VP. Type check requires a usable instrument for Instrument(e,c); selectional preference makes a timetable as noticing instrument low confidence. For "student with the timetable," prefer NP attachment unless the context establishes timetable use.

Task 4 – Lexemes Senses and WSD (8 Marks)

Sentence	POS / sense
Turn on the light.	noun: illumination/device
This bag is light.	adjective: low weight
The light meal was served early.	adjective: not heavy/rich

Use target POS, syntactic head and context. Electrical words {turn on,switch,bulb} select ILLUMINATION; physical-object heads {bag,suitcase} select LOW_WEIGHT; food heads {meal,lunch} select NOT_RICH. Return sense and evidence; retain ties. "The light suitcase fit overhead" is LOW_WEIGHT. "A light course" may mean easy or low-calorie, so domain data or clarification is needed.

For eight labelled items (3 illumination, 3 low-weight, 2 not-rich), 7 correct gives accuracy=7/8=87.5%. Label genuinely unresolved cases AMBIGUOUS. Class imbalance can hide failure on rare senses.

Task 5 – Semantic Information Retrieval (8 Marks)

Intent: identify intended word meaning from context. Score=0.5 semantic similarity + 0.3 lexical overlap + 0.2 concept coverage.

Doc	Description	Semantic	Lexical	Coverage	Score	Relevant
D1	Contextual word-sense choice	0.95	0.40	1.00	0.795	Yes
D2	Supervised sense classifiers	0.89	0.31	0.90	0.718	Yes
D3	Syntactic attachment ambiguity	0.58	0.48	0.45	0.525	No
D4	Dictionary definitions	0.42	0.55	0.35	0.445	No
D5	Contextual embeddings for polysemy	0.91	0.18	0.95	0.700	Yes

Ranking is D1, D2, D5, D3, D4; D5 beats lexical distractors through semantic coverage. At cutoff 3, precision=3/3=1.00; if there are four relevant documents, recall=3/4=0.75. D3 is a false positive and a missed relevant document is a false negative. Add a WSD task classifier and synonym expansion.

Task 6 – Integrated Semantic Analysis Challenge (8 Marks)

Input: "I watched the crane by the bank, then it moved."

Candidate	Representation
A: bird + riverbank	Watch(e,Speaker,c) ∧ CraneBird(c) ∧ Near(c,b) ∧ BankRiver(b) ∧ Moved(c).
B: machine + financial bank	Watch(e,Speaker,c) ∧ CraneMachine(c) ∧ Near(c,b) ∧

	BankFinancial(b) ∧ Moved(c).

Water, feathers, nesting and river favour A; construction, operator, load and branch favour B. Coreference “it” must agree with the selected crane entity. Retain both hypotheses and ask “Was the crane a bird or a machine?” when evidence is insufficient.

Document	**Compatibility**	**Rank**
D1 Waterbird movement near riverbanks	bird + riverbank + moved	1
D2 Mobile crane operation safety	machine + moved	2
D3 Crane migration patterns	bird; no riverbank	3
D4 Bank account closure	bank lexical only	4

With bird/river evidence, select D1. A phrase such as “near the branch” may wrongly select D4. Refinement: joint-sense scoring should require crane and bank senses to describe one coherent scene rather than resolving each word independently.

Appendices – Detailed Semantic Analysis Evidence

The following appendices provide the intermediate representations, finite models, alternative parses, test cases and ranked traces that support the decisions in Parts I and II. They are written as reproducible evidence rather than as source code.

Appendix A1 – Event Representation Trace: Maya Gave Ravi a Book

Worked input: “Maya gave Ravi a book yesterday.”

Structured representation

GIVE(e) ∧ Agent(e,Maya) ∧ Theme(e,b) ∧ Book(b) ∧ Recipient(e,Ravi) ∧ Time(e,yesterday)

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	A valid transfer event must have one Agent, one Theme and one Recipient. The temporal value may be coarse, but it must modify the event rather than the book.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix A2 – Event Representation Trace: Leena Sent the Report

Worked input: “Leena sent the report to Arun after the meeting.”

Structured representation

SEND(e) ∧ Agent(e,Leena) ∧ Theme(e,r) ∧ Report(r) ∧ Recipient(e,Arun) ∧ After(e,m)

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The relation After(e,m) is directional. Reversing it says the meeting followed sending and changes the claim.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix A3 – Role Inventory and Surface Forms

Worked input: “Maya gave Ravi a book yesterday; Yesterday, a book was given to Ravi by Maya.”

Structured representation

Agent=Maya; Theme=book; Recipient=Ravi; Time=yesterday

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Active and passive syntax differ, but semantic roles remain stable. This is why dependency or voice information must be used before assigning roles.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix A4 – Role Validation Cases

Worked input: “Leena sent the report; The report sent Arun; Leena sent Arun the report.”

Structured representation

Required(SEND)={Agent,Theme,Recipient}

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The first input is incomplete for a delivery interpretation, the second is role-incompatible, and the third is a complete ditransitive frame.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix B1 – FOPC Vocabulary Audit

Worked input: "Students, courses and instructors."

Structured representation

Student(x), Course(c), Instructor(i), Enrolled(x,c), Teaches(i,c), Passed(x,c)

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Argument order is fixed: Enrolled(person,course), Teaches(instructor,course), Passed(student,course). Swapping arguments is a type error.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix B2 – Existential Scope Trace

Worked input: “Some student passed every course taught by Dr. Rao.”

Structured representation

$\exists x[\text{Student}(x) \wedge \forall c((\text{Course}(c) \wedge \text{Teaches}(\text{rao},c)) \rightarrow \text{Passed}(x,c))]$

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The student witness is selected once outside the universal course loop. This prevents a different witness for each course.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix B3 – Quantifier Reversal Countermodel

Worked input: “Every Rao course was passed by some student.”

Structured representation

$\forall c[(\text{Course}(c) \land \text{Teaches}(rao,c)) \rightarrow \exists x(\text{Student}(x) \land \text{Passed}(x,c))]$

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	With Ana passing NLP and Bala passing IR, this formula is true while $\exists x \forall c$ is false. The formulas are not equivalent.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix B4 – Consistency Table

Worked input: "Every student passed NLP; Some student did not pass NLP."

Structured representation

$\forall x(Student(x) \rightarrow Passed(x,nlp)); \exists x(Student(x) \wedge \neg Passed(x,nlp))$

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Under one course and one assessment context, the conjunction has no model. A time index could avoid conflict only if explicitly introduced.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix B5 – Course Platform Inference

Worked input: “Ana is a learner enrolled in NLP; Python is a prerequisite; Ana completed Python.”

Structured representation

Learner(ana) ∧ Enrolled(ana,nlp) ∧ Prerequisite(py,nlp) ∧ Completed(ana,py)

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Applying the eligibility rule yields Eligible(ana,nlp). The derivation is forward chaining over a finite fact set.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix C1 – PP Attachment Parse 1

Worked input: “The researcher saw the student with the telescope.”

Structured representation

[VP saw [NP the student] [PP with the telescope]]

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The PP modifies the seeing event. Its semantic contribution is Instrument(e,telescope).	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix C2 – PP Attachment Parse 2

Worked input: “The researcher saw the student with the telescope.”

Structured representation

[VP saw [NP the student [PP with the telescope]]]

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The PP restricts the student referent. Its semantic contribution is With(student,telescope).	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix C3 – Attachment Preference Evidence

Worked input: “The researcher saw the student with the telescope.”

Structured representation

Instrument likelihood: telescope+see high; possession likelihood: student+telescope plausible.

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Both parses remain viable without discourse. The system should retain two hypotheses rather than manufacture certainty.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix C4 – Camera Sentence Parse 1

Worked input: “The analyst noticed the student with the camera near the library.”

Structured representation

Notice(e,a,s) ∧ With(s,camera) ∧ Near(s,library)

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	NP attachment produces a student description. The library PP is also recorded as an attachment decision, not silently discarded.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix C5 – Camera Sentence Parse 2

Worked input: “The analyst noticed the student with the camera near the library.”

Structured representation

Notice(e,a,s) ∧ Instrument(e,camera) ∧ Near(e,library)

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	VP attachment models an analyst using a camera. Selectional preference favours instruments that are usable for noticing.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix C6 – Semantic Role Error Tests

Worked input: “The report was sent by Leena; Arun was sent the report.”

Structured representation

Passive subject=Theme in first case; Recipient in second case.

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	A subject-position-only rule would assign Agent incorrectly. Voice and predicate frame checks catch the error.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix D1 – BANK Sense Inventory

Worked input: “bank; banked; banks.”

Structured representation

BANK_N_FINANCIAL, BANK_N_RIVER_EDGE, BANK_V_AIRCRAFT_TILT

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The inventory separates word form, POS and sense. Inflection is attached to the lexeme, while sense is chosen in context.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix D2 – BANK Financial Feature Trace

Worked input: “Nila deposited cash at the bank.”

Structured representation

deposit=3; cash=3; noun POS=0; total FINANCIAL=6

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The financial candidate wins. River and aircraft candidates receive no positive evidence and are not selected.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix D3 – BANK River Feature Trace

Worked input: “They sat on the bank of the river.”

Structured representation

sat=3; river=3; noun POS=0; total RIVER_EDGE=6

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The phrase of the river is a strong local dependency feature; it outweighs any general financial prior.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix D4 – BANK Aircraft Feature Trace

Worked input: “The pilot banked the aircraft to the left.”

Structured representation

pilot=3; aircraft=3; left=3; verb POS=0; total AIRCRAFT_TILT=9

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Verb morphology and domain terms make the aircraft interpretation uniquely strong.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix D5 – LIGHT Sense Inventory

Worked input: “Turn on the light; This bag is light; The light meal was served early.”

Structured representation

LIGHT_N_ILLUMINATION; LIGHT_ADJ_LOW_WEIGHT; LIGHT_ADJ_NOT_RICH

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	POS is a first-stage restriction. Noun versus adjective prevents several otherwise plausible confusions.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix D6 – LIGHT Rule Trace

Worked input: “The light suitcase fit overhead; The light meal was served early.”

Structured representation

suitcase→LOW_WEIGHT; meal→NOT_RICH

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The head noun supplies the strongest feature. If head type is unknown, retain candidates and lower confidence.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix D7 – WSD Failure Catalogue

Worked input: “The bank was steep; a light course; the crane moved.”

Structured representation

underspecified context → multiple candidates

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Failure is an expected output state. The system records ambiguity, requests context or passes alternatives to downstream ranking.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix D8 – WSD Evaluation Worksheet

Worked input: “Twelve BANK instances; eight LIGHT instances.”

Structured representation

Accuracy=correct/total; Precision_sense=TP/(TP+FP); Recall_sense=TP/(TP+FN)

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Overall accuracy must be paired with per-sense measures because a majority sense can hide poor minority-sense performance.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix E1 – Query Intent Decomposition

Worked input: “methods for finding the meaning of ambiguous words”

Structured representation

Intent=LEARN_WSD_METHOD; concepts={WSD,context,ambiguity,method}

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Intent distinguishes this request from dictionary lookup, syntactic ambiguity, or general linguistics.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix E2 – Semantic Document Cards

Worked input: “Contextual disambiguation of polysemous terms; Dictionary of banking terms.”

Structured representation

D1={WSD,context,decision rule}; D2={bank,definitions}

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	D1 is relevant despite paraphrase. D2 shares a word domain but does not explain intended-sense selection.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix E3 – Ranking Formula Audit

Worked input: "Score(d)=0.45 lexical+0.35 semantic+0.20 coverage."

Structured representation

Signals are normalised to [0,1] before weighting.

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Normalisation prevents a high-scale lexical score from overwhelming semantic compatibility. Tie handling is deterministic.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix E4 – Ranking Trace

Worked input: "D1 contextual WSD; D2 supervised tagging; D3 parsing ambiguity; D4 dictionary; D5 embeddings."

Structured representation

D1=.795; D2=.718; D5=.700; D3=.525; D4=.445

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	D5 ranks above literal distractors because its semantic and coverage signals demonstrate that it answers the WSD task.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix E5 – Retrieval Metric Trace

Worked input: "Top five retrieved: D1,D2,D3,D4,D5. Relevant corpus set: D1,D2,D6,D7."

Structured representation

Precision@5=2/5=.40; Recall@5=2/4=.50

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The denominator for precision is retrieved documents; the denominator for recall is all relevant documents in the collection.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix E6 – False Positive Review

Worked input: “A document about PP attachment ambiguity is retrieved for a WSD question.”

Structured representation

relatedness high; answerability low

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Semantic relatedness alone is insufficient. Add a task-type or answerability feature to reduce this error.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix F1 – DUCK and BANK Candidates

Worked input: “I saw her duck near the bank, and it was closed.”

Structured representation

Candidate A: DuckBird+BankFinancial; Candidate B: DuckAction+BankRiver

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Possessive noun versus object-pronoun-plus-verb creates a syntactic ambiguity. Bank and it add lexical/coreference ambiguity.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix F2 – DUCK Coreference Check

Worked input: "…near the bank, and it was closed."

Structured representation

Closed(bank) preferred over Closed(duck)

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Closed selects an organisation or venue more naturally than an animal. This is a probabilistic preference, not a grammar rule.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix F3 – CRANE and BANK Candidates

Worked input: “I watched the crane by the bank, then it moved.”

Structured representation

CraneBird+BankRiver; CraneMachine+BankFinancial

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Both candidates allow it=crane. Scene coherence is required to rank the joint interpretation.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix F4 – Joint Sense Compatibility

Worked input: “water, feathers, nesting; construction, operator, load, branch.”

Structured representation

Score(scene)=sense evidence + role compatibility + document match

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Joint scoring prevents selecting a bird crane with a financial-bank article merely because each word was resolved independently.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix F5 – Clarification Policy

Worked input: “Was the crane a bird or a machine? Do you mean a bank branch?”

Structured representation

Ask when best-candidate margin < threshold τ.

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	A clarification question is preferred when incorrect automation is more costly than a short dialogue turn.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix G1 – Representation Comparison Matrix

Worked input: “Event frames, FOPC, graphs and vectors.”

Structured representation

Frame=queryable roles; FOPC=truth conditions; graph=relations; vector=similarity

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	No single representation is sufficient for every task. The pipeline uses the representation that matches the decision being made.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix G2 – Symbol and Predicate Checklist

Worked input: “Every formula in this report.”

Structured representation

Type predicates; constants; variable domains; argument order; scope.

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	A formula is accepted only after each nonlogical symbol has an intended meaning. This prevents visually correct but uninterpretable notation.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix G3 – Assumption Register

Worked input: “Distinct names; fixed course identity; local context window; finite document collection.”

Structured representation

Assumptions are explicit metadata, not hidden facts.

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	When an assumption fails, the corresponding inference is withdrawn or marked low confidence.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix G4 – Error Handling Workflow

Worked input: "Input → candidate parses → role/type checks → WSD → retrieval → evaluation."

Structured representation

retain alternatives if confidence is low

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The workflow separates ambiguity detection from resolution. A failed early check does not become a confident downstream fact.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix H1 – Worked End-to-End Trace 1

Worked input: “Maya gave Ravi a book yesterday.”

Structured representation

parse→GIVE frame→role validation→query answer Recipient=Ravi

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The final answer is traceable to an event variable and named participant slot.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix H2 – Worked End-to-End Trace 2

Worked input: “How can a system identify intended word meaning in context?”

Structured representation

intent→WSD concept vector→ranked documents→precision/recall

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	The retrieval result remains explainable because every score component is recorded.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix H3 – Worked End-to-End Trace 3

Worked input: “I watched the crane by the bank, then it moved.”

Structured representation

parse alternatives→joint WSD→coreference→sense-compatible retrieval

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	Where evidence is tied, the output contains alternatives and a clarification request rather than a fabricated answer.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

Appendix H4 – Submission Checklist

Worked input: “Input, representation, trace, test, counterexample, evaluation and interpretation.”

Structured representation

Each task maps to at least one artefact in this report.

Analysis stage	Recorded evidence	Decision / interpretation
Input normalisation	Tokens, candidate lexical entries and syntactic roles are retained.	Do not discard ambiguity before evidence is collected.
Semantic composition	Entities and events are assigned stable identifiers.	Roles are checked against the predicate or event frame.
Validation	This checklist confirms that final answers are supported by intermediate semantic evidence, not only by conclusions.	Keep the highest-supported interpretation and record alternatives when necessary.

Reasoned trace

First identify the semantic units that are explicitly present in the input. Next compose their meanings according to the selected syntactic structure, preserving attachment choices and quantifier scope. Then apply type, role and selectional-preference checks. The resulting representation is suitable for direct querying, consistency checking or retrieval because every entity, relation and uncertainty source is explicit.

The evidence above also makes the analysis reproducible. A later reader can change an assumption, rerun the selection rule, and see which conclusion changes. This is preferable to a paraphrase because it separates the observed wording from the semantic commitments inferred from it.

References

- Jurafsky, D. and Martin, J. Speech and Language Processing, semantic analysis and WSD chapters.
- Manning, C. D., Raghavan, P., and Schütze, H. Introduction to Information Retrieval.
- Russell, S. and Norvig, P. Artificial Intelligence: A Modern Approach.
- Assignment prompt: DSA03_NLP_Assignment-4.pdf, supplied by the user.
- Answer format reference: DSA03_NLP_Assignment3_Answers.docx.pdf, supplied by the user.

Document typeface: Times New Roman throughout. The editable Word file is the primary deliverable.